

## SIMUSIR, Russian Federation

WMO: 321950

Lat: 46.850N

Lon: 151.867E

Elev: 82

StdP: 14.65

Time Zone: 10.00 (E10)

Period: 86-97

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 14.5       | 16.6 | 6.0                           | 7.5 | 16.5 | 9.0 | 8.8 | 17.7 | 61.8                  | 25.9 | 53.8 | 23.6 | 22.7                  | 290  | 0.964 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.9                    | 66.5            | 62.1 | 62.2 | 58.5 | 59.0 | 56.3 | 63.2                | 65.7 | 59.6 | 61.5 | 56.7 | 58.3 | 16.4                 | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.0                            | 83.2 | 65.2 | 58.7 | 73.9 | 60.8 | 55.8 | 66.4 | 57.8 | 28.8          | 65.8 | 26.2 | 61.7 | 24.3 | 58.4 | 74.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |     |
| (4) 48.4          | 41.5 | 36.8 | DB                         | 11.5 | 75.3               | 2.9 | 3.7                                                | 9.4 | 78.0       | 7.6 | 80.1       | 6.0 | 82.2       | 3.9 | 84.9 | (4) |
| (5)               |      |      | WB                         | 10.8 | 69.7               | 3.3 | 3.3                                                | 8.5 | 72.1       | 6.6 | 74.0       | 4.8 | 75.9       | 2.4 | 78.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |                            | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|----------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |                            | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg                      | 37.8        | 24.9  | 25.0  | 27.8  | 33.8  | 38.9  | 42.0  | 46.8  | 51.6  | 50.5  | 45.3  | 36.1  | 29.8  |  |  |
|                                                                        | DBStd                      | 10.31       | 4.81  | 4.91  | 4.00  | 4.39  | 4.51  | 4.15  | 4.69  | 5.26  | 4.36  | 4.28  | 5.06  | 4.67  |  |  |
|                                                                        | HDD50                      | 4643        | 779   | 700   | 689   | 485   | 345   | 244   | 123   | 37    | 42    | 158   | 416   | 625   |  |  |
|                                                                        | HDD65                      | 9932        | 1244  | 1120  | 1154  | 935   | 809   | 690   | 563   | 416   | 434   | 611   | 866   | 1090  |  |  |
|                                                                        | CDD50                      | 188         | 0     | 0     | 0     | 0     | 1     | 4     | 25    | 88    | 59    | 11    | 0     | 0     |  |  |
|                                                                        | CDD65                      | 3           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74                      | 13          | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 9     | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80                      | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg                      | 16.8        | 22.1  | 22.2  | 20.9  | 19.0  | 14.9  | 11.9  | 10.8  | 11.4  | 13.0  | 16.6  | 18.3  | 20.5  |  |  |
|                                                                        | PrecAvg                    | 53.7        | 3.7   | 3.2   | 3.7   | 3.9   | 3.8   | 3.4   | 4.4   | 5.2   | 6.2   | 5.8   | 5.6   | 4.8   |  |  |
| (15) Precipitation                                                     | PrecMax                    | 67.2        | 5.7   | 8.0   | 7.0   | 7.1   | 5.8   | 6.1   | 8.9   | 12.0  | 10.0  | 9.2   | 11.7  | 7.5   |  |  |
|                                                                        | PrecMin                    | 41.4        | 2.0   | 1.5   | 1.8   | 1.8   | 1.6   | 1.3   | 1.4   | 1.5   | 2.8   | 2.7   | 2.6   | 3.0   |  |  |
|                                                                        | PrecStd                    | 6.6         | 1.1   | 1.3   | 1.2   | 1.3   | 1.0   | 1.1   | 1.7   | 2.3   | 1.8   | 1.6   | 1.9   | 1.1   |  |  |
|                                                                        |                            |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%                       | DB          | 36.2  | 36.2  | 40.2  | 52.4  | 61.0  | 60.3  | 68.1  | 75.2  | 72.4  | 61.3  | 54.3  | 46.1  |  |  |
|                                                                        |                            | MCWB        | 34.3  | 35.0  | 37.0  | 45.5  | 50.0  | 55.3  | 63.7  | 69.1  | 69.1  | 57.3  | 50.8  | 44.3  |  |  |
|                                                                        | 2%                         | DB          | 34.8  | 34.4  | 36.4  | 46.1  | 54.1  | 54.6  | 62.4  | 68.1  | 64.1  | 57.1  | 48.3  | 41.7  |  |  |
|                                                                        |                            | MCWB        | 33.3  | 33.1  | 34.4  | 42.0  | 47.7  | 50.6  | 59.1  | 64.5  | 59.7  | 53.6  | 46.3  | 40.5  |  |  |
|                                                                        | 5%                         | DB          | 33.7  | 33.2  | 34.5  | 41.8  | 49.9  | 50.8  | 57.5  | 63.2  | 60.0  | 54.1  | 46.0  | 39.0  |  |  |
|                                                                        |                            | MCWB        | 32.5  | 31.9  | 33.1  | 38.9  | 44.3  | 48.2  | 55.3  | 60.6  | 56.1  | 50.8  | 44.5  | 37.7  |  |  |
|                                                                        | 10%                        | DB          | 32.6  | 32.2  | 33.1  | 39.4  | 46.0  | 47.8  | 53.4  | 59.5  | 57.0  | 51.6  | 43.8  | 35.9  |  |  |
|                                                                        |                            | MCWB        | 31.2  | 31.0  | 31.8  | 37.5  | 42.6  | 46.0  | 52.0  | 57.9  | 54.3  | 49.0  | 42.4  | 34.4  |  |  |
|                                                                        | 0.4%                       | WB          | 35.0  | 35.7  | 38.1  | 46.8  | 53.2  | 56.5  | 64.1  | 69.3  | 69.0  | 57.5  | 51.6  | 44.5  |  |  |
|                                                                        |                            | MCDB        | 35.6  | 36.0  | 39.7  | 51.6  | 56.8  | 59.3  | 66.7  | 72.2  | 72.5  | 60.4  | 53.4  | 45.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%                         | WB          | 33.7  | 33.5  | 35.1  | 42.1  | 47.8  | 51.4  | 60.0  | 64.7  | 60.6  | 54.3  | 47.1  | 41.0  |  |  |
|                                                                        |                            | MCDB        | 34.3  | 34.0  | 36.1  | 45.4  | 52.7  | 53.9  | 62.2  | 67.8  | 63.0  | 56.4  | 48.2  | 41.6  |  |  |
|                                                                        | 5%                         | WB          | 32.7  | 32.3  | 33.4  | 39.7  | 44.8  | 48.6  | 55.5  | 61.0  | 57.0  | 51.7  | 44.8  | 37.6  |  |  |
|                                                                        |                            | MCDB        | 33.7  | 33.0  | 34.3  | 41.5  | 48.5  | 50.4  | 57.4  | 62.5  | 58.8  | 53.3  | 45.6  | 38.6  |  |  |
|                                                                        | 10%                        | WB          | 31.3  | 31.0  | 32.1  | 37.7  | 42.5  | 46.1  | 52.2  | 58.0  | 54.6  | 49.4  | 42.2  | 34.6  |  |  |
|                                                                        |                            | MCDB        | 32.6  | 32.0  | 32.9  | 39.0  | 45.2  | 47.3  | 53.1  | 59.4  | 56.4  | 50.9  | 43.4  | 35.7  |  |  |
|                                                                        | 5% DB                      | MDBR        | 5.5   | 4.9   | 5.3   | 6.5   | 8.9   | 8.1   | 8.3   | 9.9   | 9.3   | 8.3   | 6.8   | 5.8   |  |  |
|                                                                        |                            | MCDBR       | 5.9   | 5.3   | 7.4   | 13.1  | 18.8  | 15.1  | 16.3  | 16.6  | 14.9  | 12.5  | 10.8  | 10.2  |  |  |
|                                                                        |                            | MCWBR       | 6.1   | 5.3   | 7.4   | 9.7   | 12.1  | 11.5  | 13.2  | 13.6  | 11.3  | 10.2  | 10.9  | 10.2  |  |  |
|                                                                        |                            | MCWBR       | 6.4   | 5.4   | 7.4   | 12.8  | 15.9  | 14.3  | 15.7  | 16.2  | 14.0  | 11.9  | 10.8  | 10.0  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB                      | MCWBR       | 6.5   | 5.6   | 7.5   | 9.8   | 11.8  | 11.4  | 13.1  | 13.6  | 12.0  | 10.4  | 11.2  | 10.1  |  |  |
|                                                                        |                            |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        | Clear-Sky Solar Irradiance | taub        | 0.360 | 0.382 | 0.443 | 0.492 | 0.502 | 0.484 | 0.466 | 0.440 | 0.394 | 0.403 | 0.393 | 0.370 |  |  |
|                                                                        |                            | taud        | 2.117 | 2.076 | 1.962 | 1.894 | 1.942 | 2.074 | 2.201 | 2.285 | 2.372 | 2.168 | 2.098 | 2.066 |  |  |
| (42) All-Sky Solar Radiation                                           | RadAvg                     | Ebn at Noon | 216   | 239   | 241   | 242   | 245   | 251   | 254   | 256   | 259   | 229   | 200   | 192   |  |  |
|                                                                        |                            | Edh at Noon | 30    | 38    | 49    | 58    | 57    | 50    | 44    | 39    | 33    | 34    | 30    | 28    |  |  |
| (44) All-Sky Solar Radiation                                           | RadStd                     | RadAvg      | 344   | 561   | 920   | 1189  | 1305  | 1348  | 1294  | 1141  | 1051  | 755   | 405   | 284   |  |  |
|                                                                        |                            | RadStd      | 23    | 46    | 79    | 137   | 154   | 113   | 123   | 84    | 93    | 54    | 36    | 16    |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page